

OmniDexter - A Tendon Driven 4 DOF Wrist with Gripper

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Abstract

This paper introduces OmniDexter, a novel 3-degree-of-freedom (3-DOF) tendon-driven robotic wrist integrated with a 2-DOF underactuated gripper, powered by a single degree of actuation. The design leverages three concentric layers: the inner layer enables the gripper's open-close function via worm gears, the middle layer facilitates 360° continuous rotation of the wrist, and the outermost layer incorporates a four-pulley system, which uses a unique antiparallel link mechanism in the middle layer to achieve spherical rolling movement in the X and Y axes.

Keywords: *tendon driven; antiparallel mechanism*

1 Introduction

The challenge of developing humanoid robots that are strong, agile, and safe remains a core issue in robotics. A promising step toward addressing this issue is reflected in the work of Yong-Jae Kim, Jong-In Kim, and Wooseok Jang from Naver Labs, who developed the LIMS2-AMBIDEX (LIMS2)—an innovative device with a 3-DOF wrist mechanism. This project has been a source of inspiration for our own mechatronics project.

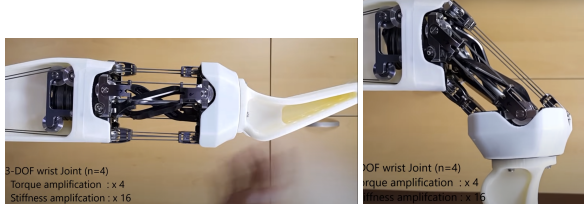


Figure 1: Screenshots of Video "LIMS2-AMBIDEX mechanical design" posted in Youtube.

LIMS2 is tendon-driven and achieves pure spherical rolling motion within a confined central space, enabled by a unique parallel link mechanism. This parallel structure's simplicity, compactness, and reliability provide fast, precise, and high-payload actuation.[Kim et al. 2018]

The LIMS2 wrist mechanism stands out for its uniform manipulation capabilities and large range of motion, all while maintaining rigidity and strength. Additionally, the tendon-driven design reduces mass and inertia and improves back-drivability, enhancing safety.

Mechanisms like this can potentially replace tedious or dangerous tasks in the future, seamlessly integrating with human tools and workspaces. Its distinctive design and enhanced functionality make it suitable for various applications, such as industrial automation, medical robotics, and space exploration.

By contrast, many similar mechanisms face challenges in balancing safety with effectiveness. For instance, while the KUKA-DLR Lightweight Robot arm boasts a low mass-to-payload ratio and programmable active compliance, its force and torque sensors are unreliable at high speeds.[Bischoff

et al. 2010] Moreover, mechanisms employing various series elastic actuators (SEAs)[Bicchi and Tonietti 2004] and variable stiffness joints (VSJs)[Petit et al. 2014] often compromise accuracy due to low joint stiffness, trading high compliance for precision.

Our team was drawn to this design precisely because of the unique advantages that LIMS2 offers over similar mechanisms.

Based on LIMS2, we aim to develop a 4-DOF structure that includes a robotic arm and a robotic hand. To achieve this, we have chosen a design that combines tendon-driven and cable-driven approaches, as it offers several benefits. First, it allows actuators to be installed remotely from the end effector, which reduces the bulk and inertia of the manipulator system.[Guist et al.] Additionally, tendon-driven robotic manipulators are more compliant than geared and direct-driven manipulators because tendons are inherently more flexible.[Guist et al.] Furthermore, tendon-driven systems provide biomimetic advantages, such as the ability to tune stiffness through co-contraction.[Guist et al.]

Nevertheless, there are still some significant drawbacks to using a tendon-driven design. One issue is a lack of precision due to wire extension, which can cause offset. Additionally, tendon-driven systems face limitations in both static and dynamic performance due to the non-negligible elasticity of the tendons. The routing systems that guide tendons from the actuator to the joint can introduce distributed friction along the tendon path, and maintaining suitable pretension is essential to avoid cable slack.[Guist et al.] Moreover, repeated high-force transmission wears down the tendon guidance, and the resulting friction typically adds nonlinearities and stochasticity.[Nazma and Mohd 2012]

Therefore, to mitigate the impact of these drawbacks, we decided to implement a specific pulley system.

During our research, we discovered that while LIMS2 enables the robotic hand to perform complex gripping tasks, it relies on motor control. Our goal is to add an additional degree of freedom with a cable-driven design to allow the robotic hand to execute complex gripping operations. At this point, we encountered an underactuated robotic gripper[Tlegenov et al. 2014], which aligns well with our requirements. This gripper uses one degree of actuation to achieve two degrees of freedom through springs and linkages. Its adaptive grasping capability allows it to closely wrap around objects of various shapes with just one DOF actuation, making it an ideal choice for our design.[Tlegenov et al. 2014]

2 Mathematical Modeling

In this section, we will discuss both 2D mathematical analysis and 3D mathematical analysis. Each small section includes the Wrist, pulley and the gripper structure. Also, we will insert several figures, equations, tables and matlab programs to help us illustrate the mathematical modeling.

2.1 2D Modeling

2.1.1 Wrist: Basic 2D Model

Fistly, let us consider a plane in figure 2(a), where made of one pair of actuating wires AC and BD . Also, A, B are in the top plane, P_T , and C, D are in the bottom plane, P_B . P_T and P_B both have length w and they are connected with two linkages, FG and EH of length l_c . EF and GH have the same length as well (w_c). Point k is not only the crossing point of two linkages, but also the touching point of two approximated spherical planes, which will be mentioned below. Additionally, the distance between two planes is h_c and we assume the two centre points are I and J .

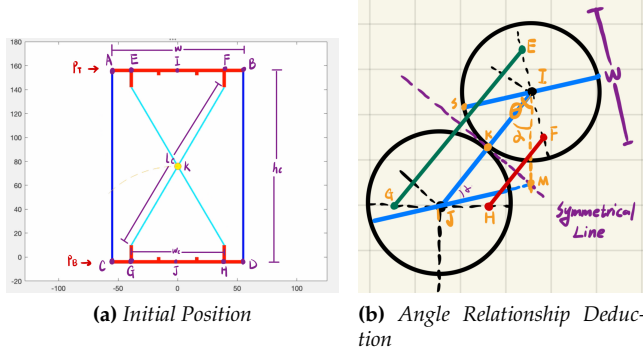


Figure 2

Figure 2(b) describes the relationship between the angle θ , the bending angle between the normal of P_T and the normal of P_B , and the angle α , which is defined as the angle between IJ and the normal of P_B . Quadrilateral $GEFH$ is always an isosceles trapezoid since $FH \parallel EG$ and the fixed length of diagonals obeying the trapezoid theorem. Therefore, as the symmetrical line crossing the point k , we could get $\angle KIM = \angle KJM = \alpha$ after making a parallel line crossing point J and extending to point M . Since two blue parallel lines, $\angle KJM = \angle SIK = \angle KIM = \alpha$. It is clear that $\angle SIK + \angle KIM = \angle SIM$, thus: Equation 1

$$\theta = 2 * \alpha \quad (1)$$

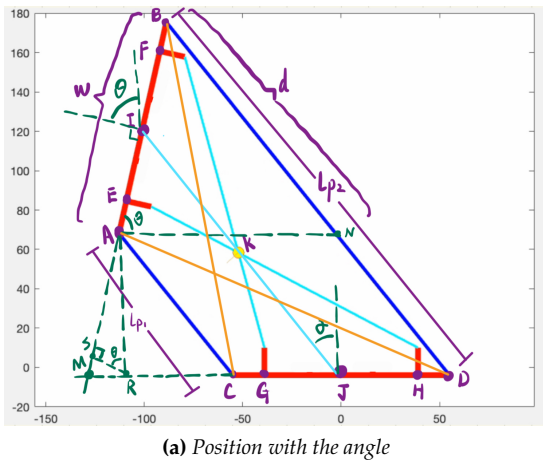


Figure 3

When the wrist twists degree θ , the difference d between l_{p1} and l_{p2} could be seen in figure 3(a). We could make two extended lines of DC and BA to intersect at point M and draw a vertical line from AN to AR and then draw a vertical line to RS as well. Therefore, $\angle SRA = \theta$. $\angle SRA + \angle SAR = 90^\circ$ and $\angle SAR + \angle BAN = 90^\circ$, thus $\angle SRA = \angle BAN = \theta$. According to the cosine rule:

$$\begin{aligned} BN^2 &= AB^2 + AN^2 - 2 * AB * AN * \cos\theta \\ d &= \sqrt{w^2 + w^2 - 2 * w * w * \cos\theta} \\ &= \sqrt{2w^2(1 - \cos2\alpha)} \\ &= \sqrt{2w^2(1 - (\cos^2\alpha - \sin^2\alpha))} \\ &= \sqrt{2w^2(\cos^2\alpha + \sin^2\alpha - \cos^2\alpha + \sin^2\alpha)} \\ &= \sqrt{4w^2\sin^2\alpha} \end{aligned} \quad (2)$$

$$\begin{aligned} d &= l_{p1} - l_{p2} = 2w\sin\alpha \\ l &= \frac{d}{2} = w\sin\theta = w\sin\frac{\theta}{2} \end{aligned} \quad (3)$$

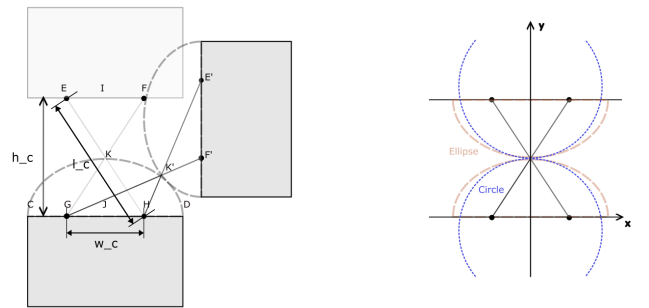
Then, using Ptolemy's theorem ($l_c^2 = l_{p1} * l_{p2} + w^2$) and Equation 3, we could get:

$$l_{pi} = \begin{cases} \pm \sqrt{l_c^2 - w^2 \cos^2 \frac{\theta}{2}} - w \sin \frac{\theta}{2}, & \text{when } i = 1 \\ \pm \sqrt{l_c^2 - w^2 \cos^2 \frac{\theta}{2}} + w \sin \frac{\theta}{2}, & \text{when } i = 2 \end{cases} \quad (4)$$

Therefore, we have found the relationship between the bending angle θ and the length of actuating wires l_{pi} .

2.1.2 Wrist: 2D Approximation

In the following section, we have already proved $GEFH$ is always a isosceles trapezoid, then we could know $GK + KH = EK + KF = \text{constant}$. The definition of ellipse is the locus of points in a plane, the sum of whose distances from two fixed points is a constant value.



(a) 2D Approximation Wrist

Figure 4

Therefore, the foci of the ellipse here are G, H and F, E and the movements of the wrist mechanism could be regarded as two ellipse movements. According to the equation of an ellipse, $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, which a represents the semi-major axis

and b represents the semi-minor axis. In figure 4(a), $b = \frac{h_c}{2}$ and $a = JD = \frac{l_c}{2}$, since $GK + KH = \text{constant} = GD + HD = JH + JD + HD = 2JD$. Then, the ellipse equation is as follows:

$$\frac{x^2}{(\frac{l_c}{2})^2} + \frac{y^2}{(\frac{h_c}{2})^2} = 1 \quad (5)$$

We could assume there is a circle which is could be approximated to the ellipse and the coordinate of circle of this circle p_o is $(0, -h_o)$, which h_o is the offset from the origin. Plus, the angle between p_c , a point on ellipse with (x_c, y_c) , and the y-axis is the angle ψ . Then, we could get:

$$\begin{aligned} y_c &= \frac{x_c}{\tan\psi} - h_o \\ r(\psi) &= \frac{x_c}{\sin\psi} \end{aligned} \quad (6)$$

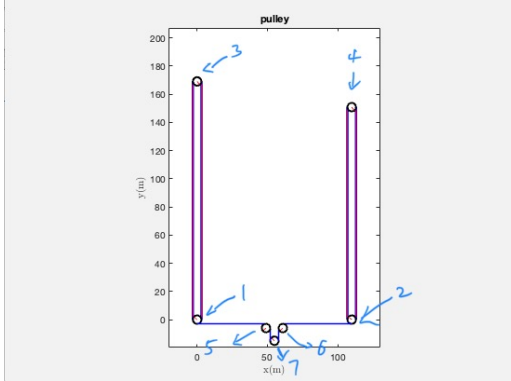
Therefore, in theory, when offset h_o is small enough and within the margin of the error, the ellipse movement of wrist could be regarded as circular motion. Substituting x_c and y_c in Equation 6 to Equation 5, r could be represented by a function of ψ :

$$r(\psi) = \frac{h_o + \sqrt{h_o^2 + (1 + \frac{w_c^2}{h_c^2} \tan^2(\psi))(\frac{w_c^2}{4} - h_o^2)}}{\cos(\psi)(1 + \frac{w_c^2}{h_c^2} \tan^2(\psi))} \quad (7)$$

In this function, it is clear to show that r changes with the angle ψ . If the range of the error is small enough with different values of ψ , r could be regarded as a constant, and thus the approximated circular motion is proved. It means that if we calculate appropriate parameters for the wrist design, this anti-parallelgram mechanism could have a high accuracy.

2.1.3 Pulley:2D

To achieve the desired wrist motion, we utilized a custom-designed pulley system called the Lightweight Tension Amplifying System.



(a) 2d model of Lightweight Tension Amplifying System

Figure 5

As shown in Figure 5(a), pulley 1 and pulley 3, as well as pulley 2 and pulley 4, each form two identical pulley blocks, but the movable pulleys always move in opposite directions. Here, pulley 1 and pulley 2 serve as fixed pulleys that can rotate, while pulley 3 and pulley 4 act as rotating movable pulleys. Within each block, four parallel lines are used between

the movable and fixed pulleys, each line wrapping around the pulleys twice, creating two parallel lines on each side of the pulley. This configuration enables the movable pulleys within each block, i.e., pulley 3 and pulley 4, to move up and down.

Therefore, we need to investigate the relationship between the change of wire L in the line between pulley 1 and pulley 5 and the corresponding vertical displacement x of pulley 3 due to L . Since there are four parallel lines in this block, we can conclude that when pulley 1 moves by X , L should change by $4X$. The equation is as follows:

$$x = 4L \quad (8)$$

Additionally, we need to determine the angular displacement of the fixed and movable pulleys within each block as L changes. For the block containing pulley 1 and pulley 3, we will start with pulley 1. Since it is a fixed pulley, when it rotates by an angle of 2π , L should change by a length of $2\pi r_1$, where r_1 is the radius of pulley 1. Thus, we get the following equation:

$$\theta_1 = \frac{L}{2\pi r_1} \cdot 2\pi \quad (9)$$

Simplifying gives:

$$\theta_1 = \frac{L}{r_1} \quad (10)$$

For pulley 3, since it is a movable pulley, when it rotates by 2π , L should change by $2\pi r_3 \cdot 4$, where r_3 is the radius of pulley 3. Therefore, we obtain:

$$\theta_3 = \frac{L}{2\pi r_3 \cdot 4} \cdot 2\pi \quad (11)$$

Simplified, this becomes:

$$\theta_3 = \frac{L}{4r_3} \quad (12)$$

Similarly, since the lines in the system of pulley 2 and pulley 4 always maintain the same motion direction as those in the system of pulley 1 and pulley 3, and the system of pulley 2 and pulley 4 are mirrors of the system of pulley 1 and pulley 3 on a line parallel to the y-axis, their pulleys will rotate in the same direction as the other set. Hence, the angular displacement of pulley 2 and pulley 4 is:

$$\begin{aligned} \theta_2 &= \frac{L}{r_2} \\ \theta_4 &= \frac{L}{4r_4} \end{aligned} \quad (13)$$

where r_2 is the radius of pulley 2, and r_4 is the radius of pulley 4.

Next, we examine the angular displacement of the three fixed pulleys, pulley 5, pulley 6, and pulley 7. Since these are also fixed pulleys, their angular displacement in magnitude aligns with the formulas derived for pulley 1 and pulley 3. In terms of direction, since pulley 5 and pulley 6 are also mirrors of each other on the line parallel to the y-axis, and the lines on them move in the same direction, their rotation directions remain the same. However, since they pull in the same direction as pulley 1 and pulley 3 but are not mirrors on the line parallel to the y-axis, their rotation direction will be opposite

to pulley 1 and pulley 3. Therefore, the complete formulas for pulley 5 and pulley 6 are:

$$\theta_{5,6} = -\frac{L}{r_{5,6}} \quad (14)$$

where r_5 is the radius of pulley 5, and r_6 is the radius of pulley 6.

Finally, since the line on pulley 7 moves in the same direction as that on pulley 5 and pulley 6, but pulley 7 is not a mirror of pulley 5 and pulley 6, the rotation direction of pulley 7 will be opposite to that of pulley 5 and pulley 6. Thus, we have:

$$\theta_7 = \frac{L}{r_7} \quad (15)$$

where r_7 is the radius of pulley 7.

2.1.4 Gripper: 2D

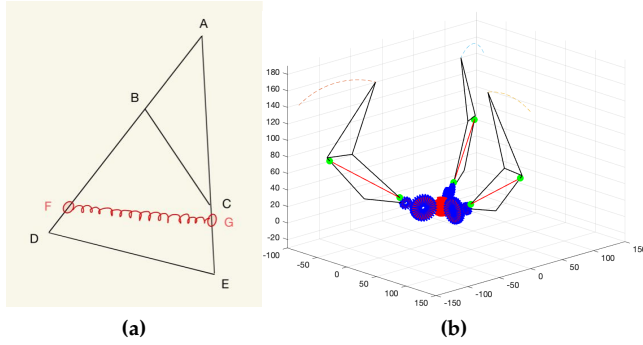


Figure 6

As shown in Figure (6), the underactuated gripper finger has 2 Degrees of Freedom but only one Degree of Actuation. The rotation of the worm drive in the z-axis will drive the first helical gear to rotate in the x-axis. By design, the helical gear has 30 teeth, module 1. The angular speed ratio between the worm drive and the helical gear would be 30:1. This is because when the worm drive rotates 1 full cycle, the helical gear only rotates $\frac{1}{30}$ cycle. The 30 teeth helical gear then drives the 13 teeth helical gear with a ratio of 30:13. The D vertex of linkage DE is fixed on the 13 teeth helical gear so that only the linkage DE will follow the rotation of the helical gear. The spring FG provides moments to linkages BD and CE to maintain the gripper shape when there is no object to handle so that the whole gripper finger will follow the movement of linkage DE in the meantime.

When a rigid object is being handled by the gripper, the finger will deform. Linkage DB touches the object first and stops rotation. Linkage AB will continue its rotation about point B driven by linkage DE through linkage CE. The angle of DEC, BCE, and DBC can be calculated by the Cosine rule when the linkage DB is fixed.

$$\cos\theta = \frac{a^2 + b^2 - c^2}{2ab} \quad (16)$$

and

$$a^2 = b^2 + c^2 - 2bc \cdot \cos A \quad (17)$$

These are achieved by finding the length of BE first by using the Cosine rule for length $BE^2 = DB^2 + DE^2 - 2DBDE \cdot$

$\cos BDE$ and then using the Cosine rule for angle to find those angles.

$$\angle DEC = \cos^{-1} \left(\frac{DE^2 + BE^2 - BD^2}{2 \cdot DE \cdot BE} \right) + \cos^{-1} \left(\frac{CE^2 + BE^2 - BC^2}{2 \cdot CE \cdot BE} \right) \quad (18)$$

$$\angle DBC = \cos^{-1} \left(\frac{BD^2 + BE^2 - DE^2}{2 \cdot BD \cdot BE} \right) + \cos^{-1} \left(\frac{BC^2 + BE^2 - CE^2}{2 \cdot BC \cdot BE} \right) \quad (19)$$

and

$$\angle BCE = \pi - \angle ABC - \angle DEC \quad (20)$$

Then the angle of DBA can be calculated as the output through input angle BDE.

$$\angle DBA = 2\pi - \angle DBC - \cos^{-1} \left(\frac{BC^2 + AB^2 - AC^2}{2 \cdot BC \cdot AB} \right) \quad (21)$$

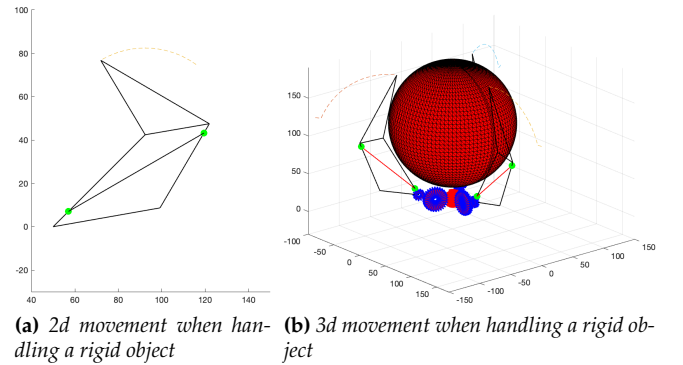


Figure 7

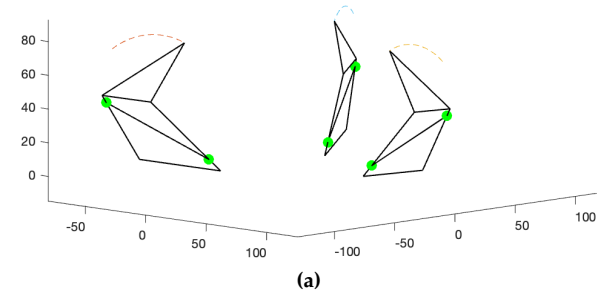


Figure 8

2.2 3D Modeling

2.2.1 Wrist: Basic 3D Model

As shown in Equation 3, l , the extension of length for one of pair of wires, equals to $wsin\frac{\theta}{2}$. However, wires are placed around the fixed pulleys and movable pulleys for multiples laps, which amplify the force on wires. As we know, the force

in pulley system is proportional to the number of windings, therefore we could get:

$$\Delta l_r = nws \sin \frac{\theta}{2} \quad (22)$$

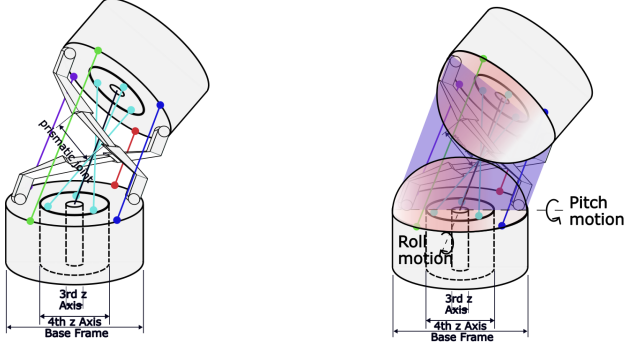


Figure 9: Frames and Bending Plane

where n denotes to the number of windings. However, in 3 dimensions, we have the 3rd degrees of freedom of z plane and we should consider the rotational angle ϕ to the mathematical analysis. In figure(9), we assume the coordinate frame p_o is (x_o, y_o, z_o) and the base frame p_e is (x_e, y_e, z_e) . In order to prove the symmetricity of pairs of wires have the same displacement Δl_r , we could consider a bending plane, where the two centerlines of the coordinate frame and base frame coexist. Meanwhile, the bending direction ϕ denote the angle between the x_o axis and the bending plane. Besides, we name one pair of wires which produces the roll motion along the x_o axis as (l_r^+, l_r^-) and another pair of wires producing the pitch motion along the y_o axis as (l_p^+, l_p^-) . This could be clearly shown in figure(5)(b).

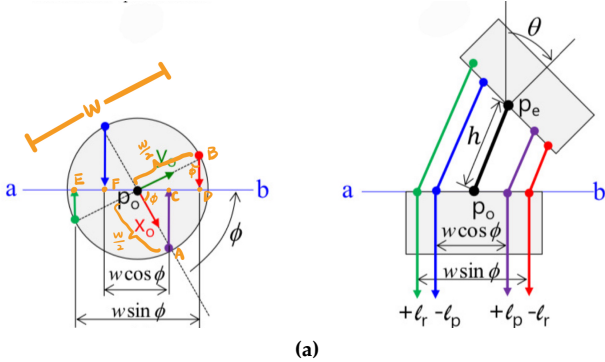


Figure 10

Figure 10 shows the top view of the rotational base frame. As we defines $\angle AP_oD = \phi$ and $AP_o = \frac{w}{2}$, we could get $P_oC = \frac{w}{2} \cos \phi$. Similarly, $\angle P_oBD = \phi$ and $P_oB = \frac{w}{2}$ concludes $DP_o = \frac{w}{2} \sin \phi$. Therefore, $CF = w \cos \phi$ and $DE = w \sin \phi$. Thus:

$$\begin{aligned} l_r &= l_r^+ = -l_r^- = nws \sin \phi \sin \frac{\theta}{2} \\ l_p &= l_p^+ = -l_p^- = nws \cos \phi \sin \frac{\theta}{2} \end{aligned} \quad (23)$$

The equations show the relationship between the angles ϕ, θ and the extension of wires l_r, l_p . It also confirms the symmetrical relationship between pairs of wires.

2.2.2 Wrist: 3D Approximation

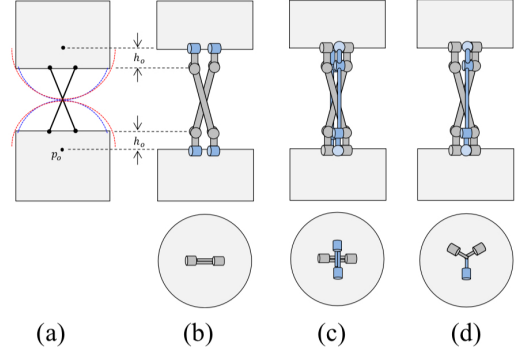
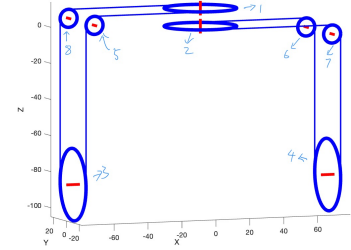


Figure 11

As shown in figure (11)(a)(b), blue parts are revolute joints added at the end of each links with offset h_o and it is the diagram of 2D Wrist. As for the 3D Wrist, the initial thought was adding another pair of links perpendicular to the original pair. However, this is a 2 DOF anti-parallelogram mechanism and it should rotate without too much constraints. So, we reallocate the positions of links and reduces the number of links to three, as shown in figure (11)(c)(d) in order to maximize the flexibility and strengthen its accuracy.

2.2.3 Pulley: 3D



(a) 3d model of Lightweight Tension Amplifying System

Figure 12

In 3D, as shown in Figure 12(a), to represent the rotation of each pulley, we chose to use different rotation matrices. First, for the driven pulleys 5, 6, 7, and 8, since they rotate around the y -axis, we use the rotation matrix for rotation around the y -axis:

$$R_y(\theta) = \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix} \quad (24)$$

For pulley 1 and pulley 2, since they rotate around the z -axis, we use the rotation matrix for rotation around the z -axis:

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (25)$$

Finally, for pulley 3 and pulley 4, which rotate around the x -axis, we can use the rotation matrix for rotation around the

x-axis:

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{bmatrix} \quad (26)$$

After discussing the rotation angle, since we wrapped one and a half turns of wire around pulleys 1, 2, 3, and 4 in Figure 12(a) to increase friction, the entire system can achieve an anti-slip effect. Therefore, we need to calculate the ratio of output force to input force to determine how much the friction on each pulley is amplified by wrapping a specific number of turns. We use the following formula to derive the result:

$$\frac{F_{out}}{F_{in}} = e^{u \cdot \theta} \quad (27)$$

where theta is the angle of the wire wrapped around the pulley, and u is the coefficient of friction.

To calculate amplification factors, We already know that the value of theta should correspond to an angle of one and a half turns, which is the length of the wire wrapped around the pulley. So, θ should be $2 \cdot \pi \cdot 1.5$. We also know that the static friction coefficient is 0.35. By substituting theta and u into the formula, we calculate the amplification factors for the pulley when stationary and rotating as follows:

$$e^{2\pi \times 1.5 \times 0.35} = 27.07666 \quad (28)$$

2.3 Iterations

Our project is a nested structure of 6 linkages connected the top platform and the bottom platform. Therefore, there are two approximated anti-parallelogram mechanism in it and it has two approximated circular motion — the inner circle and the outer circle. It is essential for us to check the feasibility of this nested structure since it doubles the number of linkages and then more constraints occurred as well. At first, we should calculate general form of radius r . r is determined by the curvature and the equation of curvature is:

$$\kappa = \frac{|x'y'' - y'x''|}{(x'^2 + y'^2)^{\frac{3}{2}}} \quad (29)$$

where κ denotes to the curvature. To be explained, the first derivative $r' = (x', y')$ gives the tangent vector at a point on the curve, and the second derivative $r'' = (x'', y'')$ represents the rate of change of the tangent vector. By the definition of the radius of curvature ($R = 1/\kappa$), we could obtain:

$$R = \frac{(x'^2 + y'^2)^{\frac{3}{2}}}{|x'y'' - y'x''|} \quad (30)$$

According to the parametric equations of the ellipse, we could obtain the coordinate of the point on the ellipse:

$$\begin{cases} x = a \cos \theta \\ y = b \sin \theta \end{cases} \quad (31)$$

Where a denotes the semi-major axis, b denotes the semi-minor axis and θ denotes the angle between the point and the x-axis, thus substituting $a = \frac{l_c}{2}$, $b = \frac{h_c}{2}$ and ψ into the equation:

$$(x, y) = \left(\frac{l_c}{2} \sin \psi, \frac{h_c}{2} \cos \psi \right) \quad (32)$$

Substituting the coordinate of (x, y) into Equation 30:

$$R = \frac{(l_c^2 \cos^2 \psi + h_c^2 \sin^2 \psi)^{\frac{3}{2}}}{2l_c h_c} \quad (33)$$

It is noteworthy that angle ψ is not a constant, we should find the mean value of R within the range of ψ for approximating the most appropriate ellipse motion:

$$R_{mean} = \frac{1}{\psi_1 - \psi_2} \int_{\psi_1}^{\psi_2} R d\psi \quad (34)$$

The range of ψ is from $\frac{\pi}{4}$ to $\frac{3\pi}{4}$ for both two structures. The offset h_o is the difference between R and R_{mean} . When we substitute the numerical values into Equation 33 and Equation 34, we obtain:

Table 1: Experimental data for offset

Base's Radius(mm)	R	R_{mean}	h_o
Inner - 12.2	63	65.9613	2.9613
Outer - 39	53	80.7078	27.7078

We have found the inner structure could be approximated successfully to the ellipse motion, while the outer structure has large value of the offset and it occurs much more constraints of our initial thoughts. Therefore, we decided to change the design of the outer structure. If we do not add three linkages to connect the outer circular top platform with the outer circular bottom platform, the accuracy will be affected seriously. To be specific, when we fix the bottom platform, rotating the top platform will form angle errors.

Therefore, we attempted to change three linkages to two linkages. The original three linkages supports 120° respectively. As for the two linkages, each one should support 180° . This could not be achieved with a rigid linkage. So, we design a prismatic joint at the center of the linkage to support it could move translational motion with the movements of whole structure.

Listing 1: matlab for R_{mean}

```
% Define glf1 and r as functions of xly1
% xly1 = hc/2, glf1 = lc/2
r = @(phi) ((glf1^2*cos(phi).^2+xly1^2*sin(phi).^2).^ (3/2)) / (2*xly1*glf1);

% Define q as a function of xly1
q = @(xly1) integral(@(phi) r(phi, xly1), -pi/4, pi/4) * 1/(pi/2);

% Range of xly1 values to test
xly1_values = linspace(10, 200, 100); % Adjust range as needed
q_values = zeros(size(xly1_values));

% Calculate q for each xly1
for i = 1:length(xly1_values)
    xly1 = xly1_values(i);
    q_values(i) = q(xly1);
end
```

Listing 2: matlab for linkage range

```
%calculate the range of the linkage
GJ = 39;
IJ = 126;
JZ = sqrt((xy^2)/2);
linkage_min = sqrt((JZ-GJ)^2 + (JZ+GJ)^2);
linkage_max = sqrt((2*GJ)^2+IJ^2);
range = linkage_max - linkage_min
```

To visualize the change of R_{mean} , which labels q in the following diagrams, we use matlab, which could be seen in listing1, to draw several graphs. Additionally, we even calculate the range of the prismatic joint length as shown in listing1.

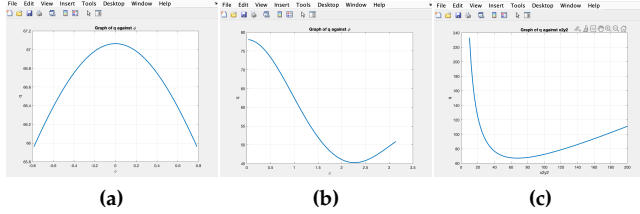


Figure 13

3 CAD Process

3.1 Wrist Iteration: Stage 1

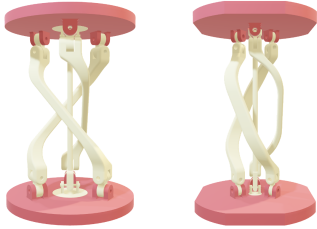


Figure 14: Wrist: Stage 1

The initial iteration, shown in the left CAD model (Figure 14), closely mimics the structure of the LIMS2 wrist as we tried to understand the structure clearly. Since the unscanned data used, the space for the shaft movements was not sufficient, which means constraints for z-axis were more serious than we thought. Therefore, we reduced the diameter of both top and bottom diameters and meanwhile made positions of three linkages more compact so that provided more space for the shaft in the second CAD model, which is the right one. Through experiments, this iteration could satisfy 45° for all curving directions successfully.

3.2 Wrist Iteration: Stage 2

After modelling first two iterations, we wanted to add some creative ideas to improve this structure. Since LIMS2 only has 3 DOF, the claw is connected directly with the wrist, thus they have the same movements for the yaw motion. What we want to achieve is separating two parts and making them independently by adding a 4th DOF for the gripper motion. In this case, the claw could not only rotate, but also carry something as the gripper by worm driven structure. The first attempt, the fourth iteration shown in Figure 15

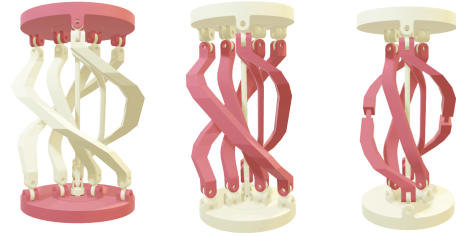


Figure 15: Wrist: Stage 2

leftside, was trying to double the original structure and design a nested structure. The outer structure was connected with the inner one with the bearing so as to movable inner structure. However, we found this structure had too much constraints even we have already reduced the diameters of the whole part in the fifth iteration, as shown in the middle one in Figure 15. To fix this problem, we attempted to reduce the three linkages to two linkages for the outer part. Two linkages could provide wider space for the movements of inner structure, while the movements of themselves were constrained since each linkage had to respond for 180° , which means larger strains exerting on linkages. As a result, we designed these two linkages with two prismatic joints at the middle. This structure could release strains on them and make them adjustable for the length while inner structure moving. This is the design of the fifth iteration, which could be seen in the right model in Figure 15.

3.3 Gripper Iteration

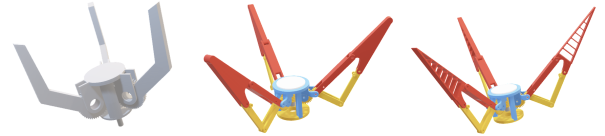


Figure 16: Gripper

As mentioned before, the 4th DOF could achieve the gripper motion using the worm driven structure. For the first iteration of gripper shown in the left in Figure 16, we designed a simple structure to test the feasibility of worm gear. Fortunately, this structure could be achieved as a foundation for our thoughts: Worm wheels move with the worm and then the gripper connected with worm wheels could achieve grasping motion. As for the second iteration of gripper, next to the first one in Figure 16, we decided to use an underactuated gripper. Underactuated gripper has higher flexibility when handling objects as springs added to the gripper part. Spring as a damper could adjust the force when grasping objects and adapting different objects with different shapes. After searching on the internet, we found there is a mechanics could adapt out projects properly: Festo's adaptive gripper. Compared with the rigid grippers as first two iterations, this gripper could allow even higher flexibility when handling objects with different shapes. Since this gripper uses the soft material, it could joint the surface of the object more closely and larger surfaces touched could provide larger frictions as well.

3.4 Pulley System Iteration

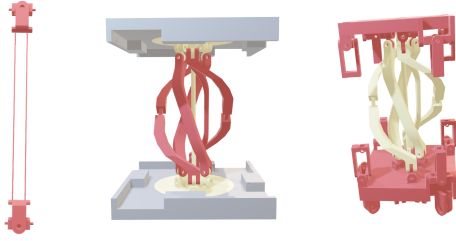


Figure 17: Pulley System

The first diagram in Figure 17 is the basic pulley design for this project. In order to achieve the reduced force structure, pulleys are all movable pulley, so that force exerting on the rope could contribute to double force for one ring. Then, for the first version of pulley system design, we designed four space for each edge for the pulley.

However, this pulley loaded structure was too large and there was no space for the 3rd and 4th Dof. In terms of the next version, we adjusted positions of four pulley space to have some degree and reduced the size of it.

4 Mechanical Analysis

4.1 4-DOF Model

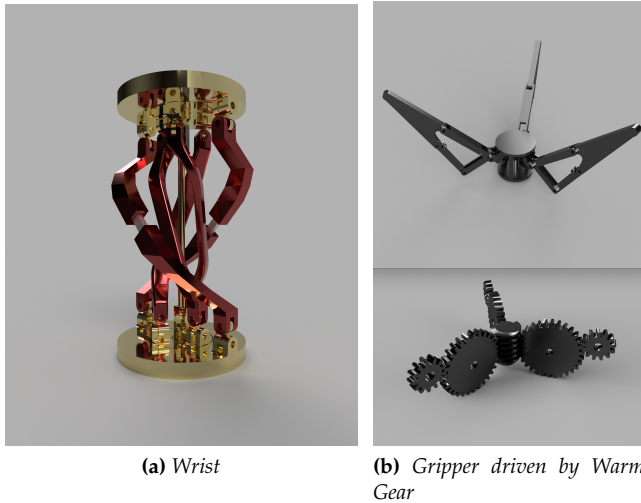


Figure 18

We now describe the wrist mechanism, as shown in Figure 18(a). This mechanism features a 4-DOF nested structure inspired by the LIMS2 design. The system consists of an inner structure with three linkages, which approximate spherical motion, and an outer structure with two additional linkages that enable a fourth degree of freedom. The outer structure was added to achieve rotation of the central shaft, which controls the opening and closing state of the gripper.

The wrist mechanism incorporates two antiparallelogram mechanisms to drive the actuation system. These mechanisms are formed by opposite pairs of the four pulleys lo-

cated on each platform. Each antiparallelogram pair ensures coordinated and precise motion.

The inner structure retains the elliptical motion characteristic of the LIMS2, successfully approximating spherical motion. However, due to larger offsets and constraints, the outer structure is different with the inner one. Specifically, the outer two linkages integrate prismatic joints, which provide flexibility to accommodate strain and allow smooth motion during wrist actuation.

In addition to the primary five linkages, the system includes 10 supporting linkages distributed between the top and bottom surfaces. These connect the five primary linkages (three for the inner structure and two for the outer structure) to the platforms, ensuring structural integrity and synchronized movement.

The actuation system employs four pulleys on each platform, with tendons driving motion through a pulley system. The antiparallelogram configuration allows the tendons to transfer forces efficiently. Further details of this tendon-driven mechanism will be described in the following subsection.

4.1.1 Pulley System

The pulley system is the core actuation method for the wrist mechanism, controlling all rotational motions, including pitch, roll, and yaw. Each platform incorporates four pulleys arranged in a clockwise pattern around the edges, positioned slightly off-center to the left. Opposite pulleys are paired, sharing a continuous tendon. When the length of one tendon decreases, the length of its paired counterpart increases, enabling angular changes in the wrist. This setup ensures synchronized and stable movement across the pitch and roll axes.

To minimize friction and cable intersection, the radii of the pulleys are carefully adjusted. This variation in radii prevents interference between the cables, allowing smoother operation.

For the yaw motion, two additional larger pulleys are located beneath the downward platform at the center. These pulleys enable motor actuation of the yaw axis, ensuring precise rotational control. Additionally, four smaller pulleys under the platform guide the tendons.

The choice to implement a tendon-driven pulley system over mechanical alternatives, such as bevel gears, was made to ensure simplicity and eliminate backlash. Furthermore, by placing the motors at the shoulder of the robotic arm, the system benefits from reduced mass at the wrist, improving safety, stability, and motion accuracy.

4.1.2 Adaptive Gripper

Featuring a system of 12 linkages, this adaptive gripper, as shown in Figure 18(b), provides an intricate balance of flexibility and stability. The gripper integrates three springs and 4x3 revolute joints, allowing for precise motion control and enabling the gripper to adapt to various objects and sizes. Its worm drive, coupled with six helical gears, enhances its ability to perform complex tasks by translating rotational motion into controlled force, ensuring optimal grip and movement.



Figure 19

4.1.3 Summary

The OmniDexter, as shown in Figure 19, contains:

- Four degrees of freedom
- Five primary linkages
- Two passive prismatic joints
- The bending angle could reach 90° in any bending plane
- Using pulley system to control all of movements (yaw, pitch and roll)
- Underactuated gripper with Festo's adaptive gripper finger

5 FEA Analysis

5.1 FEA on Wrist

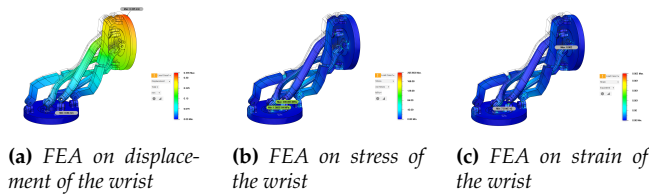


Figure 20

Table 2: Force

Type	Force
Magnitude	103.00 N
X Value	-0.28 N
Y Value	0.00 N
Z Value	-103.00 N
X Angle	90.0°
Y Angle	0.0°
Z Angle	0.0°

This is an FEA analysis of the wrist structure, which could be

seen in Figure 19. We bended the wrist to 45° at first. Then fixing the bottom platform and exerting a vertical force pointing down at the top surface. What we want to test is the maximum force that the wrist could loaded at the extreme condition. Additionally, this force will lead to how much stress, strain and displacement is also our focus as well. According to several experiments, we have tested the maximum force is 103N. This means that when the wrist bended to 45° , it could grasp $103N - \text{mass of gripper} * g$. Table 2 shows a detailed data for this.

Table 3 shows how does the force affect the stress, displacement and strain, and the minimum and maximum value for all of them:

Table 3: Result Summary

Name	Minimum	Maximum
Safety Factor:		
Safety Factor	1.006	15.00
Stress:		
von Mises	1.085E-04 MPa	205.853 MPa
1st Principal -51.726 MPa	198.243 MPa	
Normal XX	-128.155 MPa	157.18 MPa
Normal YY	-60.868 MPa	67.589 MPa
Normal ZZ	-213.527 MPa	149.623 MPa
Shear XY	-64.38 MPa	63.209 MPa
Shear YZ	-45.576 MPa	54.934 MPa
Shear ZX	-91.155 MPa	84.297 MPa
Displacement:		
Total	0.00mm	0.335mm
X	-0.009mm	0.271mm
Y	-0.021mm	0.075mm
Z	-0.216mm	0.014mm
Strain:		
Equivalent	7.514E-10	0.002
1st Principal	4.360E-10	0.002
3rd Principal	-0.001	7.495E-06
Normal XX	-4.788E-04	4.858E-04
Normal YY	-1.751E-04	1.858E-04
Normal ZZ	-9.096E-04	6.489E-04
Shear XY	-7.971E-04	7.826E-04
Shear YZ	-5.643E-04	6.801E-04
Shear ZX	-0.001	0.001

5.2 FEA on Gripper

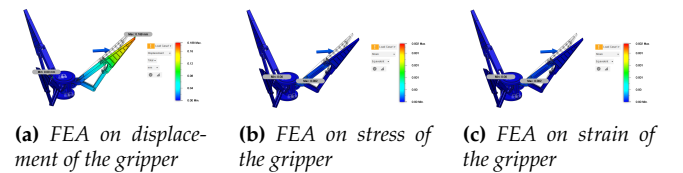


Figure 21

This is an FEA analysis of the gripper, shown in Figure 21. We want to test when the gripper located horizontally, how the vertical force exerting on the top of the gripper will change. Also, We fixed the central part of the gripper. Through several experiments, we have found that the maximum force is 11N, (Table 4) which means the maximum weight that the gripper could hold is 1.12kg. Beyond this weight, the safety factor will lower than the standard value '1'.

Table 4: Force

Type	Force
Magnitude	11.00 N
X Value	11.00 N
Y Value	0.00 N
Z Value	0.00 N
X Angle	45.0 °
Y Angle	0.0 °
Z Angle	0.0 °

However, we failed to add the spring on the gripper since there is not appropriate material to choose. We are using the Zinc Plated Spring for our model. But we cannot find the zinc material for this one. Then, we move to the steel AISI 6150, a chromium vanadium spring steel that's suitable for oil hardening and tempering. It's tough, shock resistant, and has good wear and abrasion resistance, and steel AISI 5160, a strong spring steel that's often used in applications with smaller cross-sections under significant stress, such as springs, fasteners, and leaf springs in automobile suspensions. Fusion 360 only supports the steel AISI 5160, but it is still hard to visualize the real stress and strain on springs for this model.

Table 5 shows how does the force affect the stress, displacement and strain:

Table 5: Result Summary

Name	Minimum	Maximum
Safety Factor:		
Safety Factor	1.051	15.00
Stress:		
von Mises	0.00 MPa	197.014 MPa
1st Principal -3.207 MPa	118.424 MPa	
Normal XX	-38.919 MPa	45.438 MPa
Normal YY	-9.17 MPa	11.256 MPa
Normal ZZ	-34.184 MPa	32.513 MPa
Shear XY	-103.969 MPa	103.895 MPa
Shear YZ	-98.741 MPa	97.483 MPa
Shear ZX	-34.858 MPa	30.924 MPa
Displacement:		
Total	0.00mm	0.188mm
X	-0.016mm	0.135mm
Y	-0.006mm	2.328E -04mm
Z	-0.13mm	0.002mm
Strain:		
Equivalent	0.00	0.002
1st Principal	-5.430E-09	0.001
3rd Principal	-0.001	1.281E-09
Normal XX	-2.182E-04	2.404E-04
Normal YY	-3.551E-05	4.005E-05
Normal ZZ	-2.046E-04	1.962E-04
Shear XY	-0.001	0.001
Shear YZ	-0.001	0.001
Shear ZX	-4.316E-04	3.829E-04

6 Manufacturing Methods

During the manufacturing process, we utilized four primary methods to produce our components: 3D printing, laser cutting, manual fabrication, and the acquisition of off-the-shelf parts. By applying different methods to different components, we not only saved time and cost but also ensured the

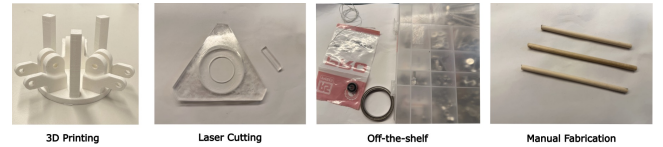
stability and precision of the final product.

Among these methods, 3D printing was the most frequently used. 3D printing enabled us to quickly and easily manufacture components with multi-dimensional and complex structures, which would be challenging to achieve using other methods. Additionally, the ability to print multiple parts simultaneously significantly accelerated the iterative process. As a result, the wrist, most of the gripper, and parts of the pulley system were manufactured using 3D printing. These components required extensive iterations and intricate designs, which made 3D printing the ideal choice.

The second most commonly used method was purchasing or borrowing off-the-shelf parts. This included all bearings, joints, screws, wires, and springs, which were standard, non-customized components. While this approach incurred higher costs, the advantages of reusability, convenience, and durability perfectly compensated for its drawbacks, making it an ideal solution for our needs.

Laser cutting was the least utilized method in our project. Despite its high precision, it was unsuitable for producing multi-dimensional or complex objects, which limited its application. Therefore, we only used laser cutting for specific parts of the gripper, specifically for the components that connect the controlling mechanism to the claw and the bearing cap used in the control section. Both of these parts demanded high precision and structural stability, and laser cutting was the perfect choice to meet these requirements.

Finally, we employed manual fabrication using existing wooden materials to produce the wooden stick in the center of the wrist. This method was highly cost-effective and offered excellent reusability. To accommodate size variations in the wrist, we simply cut away unnecessary sections, allowing the wooden stick to adapt easily to different iterations.

**Figure 22**

7 Assembly

7.1 Assembly Steps

In this section, we will include the key steps required to assemble the Omni-Dexter.

7.1.1 4 Dof Wrist

The wrist section is the foundation of the Omni-Dexter and should be assembled first. It consists of an outer passive layer, a middle anti-parallelogram layer, and a central universal joint.

Initially, as illustrated in Figure 23, we need to connect the three base links to the center base using M4 x 16 bolts and M4 non-slip nuts. The assembly should begin with the inside holes, followed by the outer holes. This sequence is essential because the height of the non-slip nuts is significantly greater than the head of the M4 bolts, which affects the movement

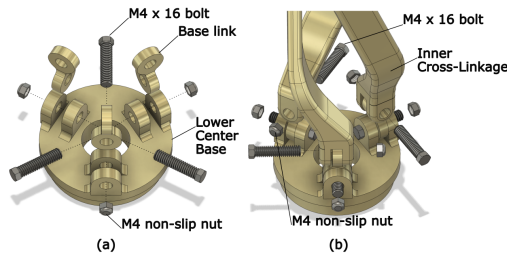


Figure 23

of the inner universal joint. These steps must be repeated twice, as there are two identical center bases—one upper and one lower. The shaft hole diameter is designed to be 4.4 mm, which has been tested to strike the right balance between being not too tight or too loose. We initially experimented with standard M4 nuts; however, due to the movement of the linkages, they frequently came loose, prompting us to switch to non-slip nuts.

Secondly, we need to connect the upper and lower center bases using the inner cross-linkage, secured with three M4 x 16 bolts and non-slip nuts on each side, as illustrated in Figure 23b and 24a. The gap offset between the inner crosslinkage is designed to be 3mm, which has been tested to ensure a precise fit based on the capabilities of the 3D printer. Sanding will be required to ensure smooth rotation.

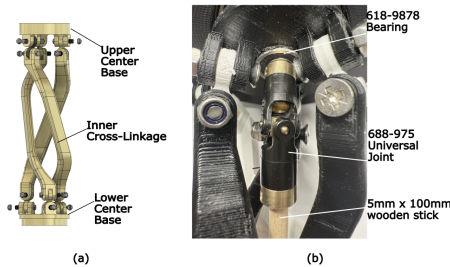


Figure 24

After assembling the middle layer of the wrist, the next step is to assemble the universal joint. This process begins by attaching two universal joints 688-975 to a 5mm x 100mm wooden shaft, which is then positioned at the center of the middle layer, as illustrated in Figure 24b. Next, it is connected via a shorter wooden shaft and 618-9878 bearings, as depicted in Figure 26b. The offset between the 618-9878 bearing and the center base is set to 3mm, a value that is relatively small, necessitating some sanding for adjustment.

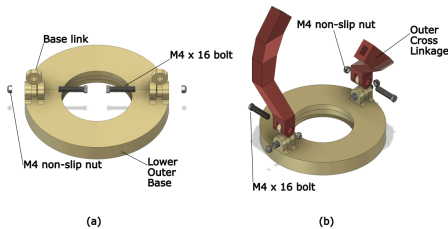


Figure 25

Next, we need to assemble the outer layer of the wrist, following the instructions depicted in Figure 25a. Start by plac-

ing two base links in the designated gaps and securing them to the outer base using two M4 x 16 bolts and M4 non-slip nuts. It is important to insert the bolts through the inner holes first, followed by the outer holes. This method is necessary because the height of the non-slip nuts exceeds that of the M4 bolt heads, which can impact the movement of the inner middle layer. Next, attach the two outer cross linkages to the base links in the same manner as illustrated in Figure 25b.

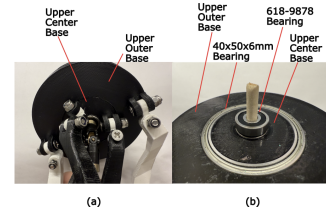


Figure 26

This process must be repeated twice, as we are working with two identical outer bases.

We will then need to assemble the two outer bases with the center bases using two 40mm x 50mm x 6mm bearings, as depicted in Figure 26a and b. These bearings serve to restrict motion along the x and y axes while permitting rotation about the z axis. Additionally, the tolerances of the 3D prints are insufficient to accommodate the bearings, so sanding will be necessary.

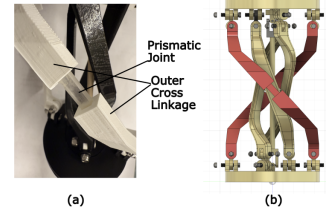


Figure 27

The final assembly process for the wrist component involves connecting the upper and lower outer Cross-Linkage using the prismatic joint, as illustrated in Figure 27a and b. The offset between the prismatic joint and the 3D prints was set to 0.2 mm. However, due to the suspended nature of the groove in the 3D prints, combined with the use of support structures, the resulting translational motion did not perform as smoothly as anticipated. Consequently, considerable sanding was required to achieve the desired fit.

7.1.2 Adaptive Gripper

After finishing assembling wrist, we need to construct the gripper part.

As illustrated in Figure 28a, begin by attaching Linkage 1 to Gear 1 using superglue. Next, connect Linkage 3 to this assembled component with an M3 x 16 bolt and M3 non-slip nut. Following that, join the two Linkage 2 pieces to Linkage 4 using an M4 x 20 bolt and M4 non-slip nut, and then secure these components together in the same manner with an M3 x 16 bolt and M3 non-slip nut. Finally, attach the spring to Linkage 3 and Linkage 2 with one M3 x 16 bolt and one M4 x 20 bolt. Once the Underactuated Proximal parts are

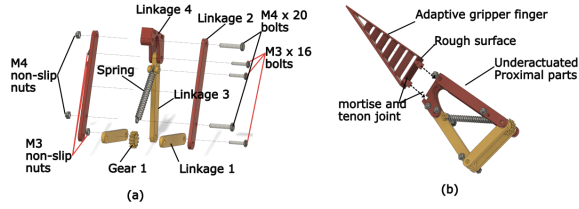


Figure 28

complete, connect the adaptive gripper finger using a mortise and tenon joint and superglue, as depicted in Figure 28b. The TPU finger features a rough surface to enhance adhesion to the PLA parts. This entire process is repeated three times, as three identical fingers are required.

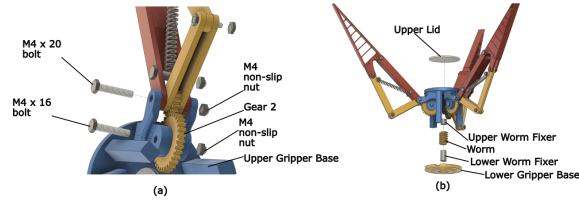


Figure 29

After assembling the adaptive fingers, begin by attaching Gear 2 to the Upper Gripper Base using an M4 x 16 bolt and an M4 non-slip nut. Next, secure the finger to the Upper Gripper with an M4 x 20 bolt and another M4 non-slip nut, as illustrated in Figure 29a. Due to an excessive offset in the shaft hole that caused significant backlash in the gripper finger, we adjusted the offset to 0.2mm to enhance performance. This process needs to be repeated three times. Finally, connect the Upper Lid, Upper Worm Fixer, Worm Drive, Lower Worm Fixer, and Lower Gripper Base using a 5mm diameter wooden shaft and a 618-9878 bearing, as depicted in Figures 29b and 26b. This assembly was relatively straightforward, as the offset was adequately aligned. Furthermore, the Worm Fixers and the Upper Lid effectively constrained the translational motion of the Worm, resulting in relatively high precision.

7.1.3 Pulley System

The last part of the Omni-Dexter is the Pulley System. It is constructed of the Lower Pulley Base and Upper Pulley Base.

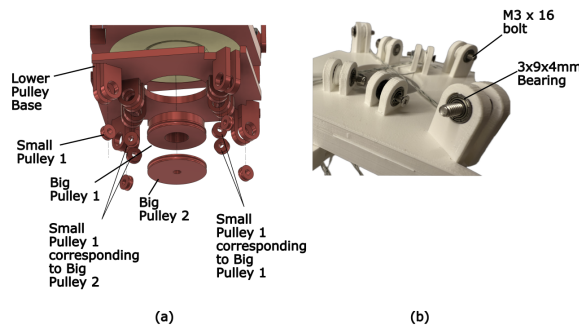


Figure 30

As illustrated in Figure 30a and b, the assembly of the Lower

Pulley Base begins by positioning the 8 Small Pulley 1 units between the pulley holders. For each pulley, secure them using two 3 x 9 x 4mm bearings and one M3 x 16 bolt. Next, attach Big Pulley 1 to the lower center base with superglue. Then, connect Big Pulley 2 to Big Pulley 1 using a 5mm diameter wooden shaft and a 618-9878 bearing, as depicted in Figure 26b.

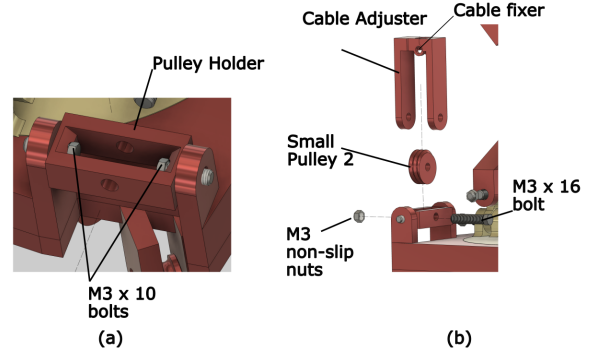


Figure 31

To assemble the Upper Pulley part, start by placing the Pulley holder between the gaps. Use two M3 x 10 bolts to secure it from the inside out, as illustrated in Figure 31a. This design ensures sufficient space for the pulley to rotate freely. Next, attach the Small Pulley 2 and Cable Adjuster to the Pulley holder using an M3 x 16 bolt and an M3 non-slip nut, as shown in Figure 31b. This process needs to be repeated eight times for the pulley base on each side.

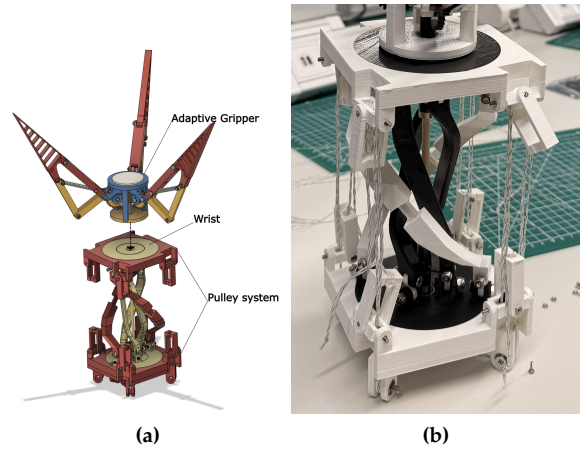


Figure 32

To finalize the assembly, we need to securely connect all three components of the Omni-Dexter using superglue, as illustrated in Figure 32. Next, we will route the cables depicted in Figure 23 through the Small Pulley 1 of Lower Pulley System Base and then up to Small Pulley 2 of Upper Pulley System Base on each side, repeating this process twice. Afterward, attach the cables to the Cable Fixer shown in Figure 31b. For the larger pulleys, guide the cables around each big pulley in two complete cycles to optimize friction before threading them through the corresponding two Small Pulley 1s depicted in Figure 30b.

7.2 Design for Assembly

During the design and iteration process, we experimented with various offset settings across different components to optimize assembly ease. For 3D prints that incorporated bolts, we found that an offset of 0.2mm provided the best balance for smooth rotational motion. In contrast, between two 3D printed parts, an offset of 0.3mm proved most effective, given that our layer height was set at 0.15mm. Furthermore, the mortise and tenon joint design for the adaptive gripper finger components was tailored to facilitate assembly, and the gap for the spring attachment was appropriately sized for a secure connection, as demonstrated in Figure 28. For the pulley holder sections, we deliberately designed a significantly larger gap than the diameter of the pulley to allow for the M3 x 10 bolts to be installed from within. A chamfer was also incorporated to ease the tightening of these bolts using screw driver, as shown in Figure 31a. However, some components, such as the prismatic joints depicted in Figure 27a, still require considerable sanding time. This issue could be alleviated by adjusting the offset for the laser-cut parts.

7.3 Potential Improvements

The prototype of the Omni-Dexter relied on several components connected with superglue, which is not ideal for an efficient assembly process. This can be enhanced by switching to more reliable connecting methods, such as mortise and tenon joints or using bolts and nuts. Additionally, 3D printing incurs significant costs in terms of both time and money, especially when sanding is required and support structures is needed to be removed. To streamline this process, molding could be employed, as it not only speed up the processes but also offers greater precision.

8 Challenge and Solution

During the manufacturing and assembly process, we encountered numerous unexpected challenges. Through brainstorming and collaborative problem-solving, we identified effective solutions and overcame these failures.

Firstly, in the early design phase, Z-axis motion lacked proper constraints, causing joints to wobble and deviate during movement. To address this issue, we introduced ball bearings, which provided precise constraint and low-friction motion. This effectively limited unnecessary degrees of freedom, enhancing both stability and linear motion precision.

Secondly, the cylindrical nylon rod initially used as the prismatic joint failed to deliver sufficient structural stability. To resolve this, we replaced it with laser-cut cuboidal acrylic sheets, which not only improved manufacturing precision but also greatly enhanced structural stability and ensured accurate motion constraints.

Additionally, during the assembly process, we observed that hole offsets were excessively large, leading to significant backlash. By adjusting the design tolerances and optimizing the manufacturing process, we successfully controlled the offset within 0.2mm, greatly reducing backlash and improving the fitting accuracy between components.

Another challenge involved material adhesion. Traditional adhesives were insufficient for bonding PLA and TPU materials, resulting in weak joints that compromised structural stability. To solve this, we adopted the mortise and tenon

joint design concept, replacing adhesive bonds with structural connections. This significantly enhanced both the connection strength and the durability of the components.

Finally, removing 3D-printed supports posed a significant challenge, particularly for helical gears, whose complex geometry required extensive supports. Traditional support materials were difficult to remove and often compromised the part's integrity. To overcome this, we used PVA water-soluble support material, which greatly improved the efficiency of support removal. Additionally, we replaced PLA with nylon material, which offered a smoother surface finish, further improving assembly precision.

9 Conclusion

The Omni-Dexter, featuring a 3-DOF wrist and a 2-DOF equipped with a 1-DOF underactuated gripper, has been successfully developed and rigorously evaluated. This innovative system demonstrates exceptional flexibility and efficiency, firmly establishing its role in the advancement of compact robotic solutions.

It utilizes tendon-driven and cable-driven mechanisms, resulting in an exceptionally lightweight design by positioning all actuators toward the main body. This arrangement significantly reduces the moment of inertia, enhancing precision and response speed, while also lowering energy consumption.

The advanced detachable design allows for the implementation of various tools, such as pliers, scissors, drills, and grinding wheels. This versatility enables the Omni-Dexter to be adapted for more complex situations.

Further refinements in the design and testing under various conditions could improve its robustness. Future research will focus on exploring different tool options, developing a quick release and assembly mechanism between the wrist and tools, enabling automatic tool changes, and investigating advanced control strategies for integration into larger robotic systems.

Overall, this work lays a strong foundation for the development of a more sophisticated and functional robotic arm platform.

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